Week 12: 16–20 November 2026

Subject / الموضوع: AP Chemistry **Unit Title / عنوان الوحدة: Unit 7 Equilibrium –** **Grade / الصف: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Session 1 الحصة الأولى 7.9 Le Châtelier's Principle مبدأ لوشاتلييه	**DOK 3 – Predict and justify:** I can predict how a system at equilibrium responds to a stress (a change in concentration, volume or temperature) and justify my prediction using Le Châtelier's principle.	3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 7.2 Make learning relevant, authentic, and connected to students' backgrounds. **Formative:** "predict the shift" cards and mini-whiteboard checks	Kindly check Toddle for the lesson PPT and practice sheet.
Session 2 الحصة الثانية Formative Assessment: ICE Tables and Le Châtelier التقييم التكويني: جداول ICE ومبدأ لوشاتلييه	**DOK 3 – Apply and justify:** I can apply ICE tables and Le Châtelier's principle to solve AP-style equilibrium problems, and justify my answers with calculations and reasoning.	7.3 Minimize threats and distractions through predictable structures and varied stimulation. 4.2 Ensure access to tools and assistive technologies. 8.4 Provide mastery-oriented feedback that emphasizes growth and improvement. **Formative:** written assessment on ICE tables and Le Châtelier's principle	Kindly check Toddle for the formative assessment and review sheet. AP Chemistry equations and constants sheet.
Session 3 الحصة الثالثة Lab: Equilibrium مختبر الاتزان	**DOK 3 – Investigate and explain:** I can apply stresses to a system at equilibrium in the lab, record my observations, and explain each change using Le Châtelier's principle.	4.1 Provide alternatives for physical action and methods of response. 8.3 Foster collaboration and community with structured group roles and supports. 6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides. **Summative project:** equilibrium lab (data collection and observations)	Kindly check Toddle for the lab sheet and safety instructions.
Session 4 الحصة الرابعة Lab Reflection Journal (Summative Project)	**DOK 3 – Evaluate:** I can evaluate my equilibrium lab in a reflection journal, using evidence from my data to explain what I learned and how I could improve my investigation.	9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	Kindly check Toddle for the reflection journal template and rubric.

دفتر التأمل في المختبر (المشروع الختامي)		5.2 Use multiple tools for composition (spellcheckers, manipulatives, templates). **Summative project:** lab reflection journal	
Homework الواجبات			
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Session 2 is a formative assessment on ICE tables and Le Châtelier's principle. Students need a calculator and the AP equations and constants sheet. Sessions 3 and 4 form the summative lab project: students carry out the equilibrium lab and then write a reflection journal. Students must wear goggles and follow the lab safety rules.		

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Technology Integration دمج التكنولوجيا	AP Classroom (AP Daily videos and Topic Questions), Toddle for lesson materials, the formative assessment and the reflection journal, and a digital lab journal for observations and photos.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics (solving for unknowns in ICE tables) and Economics (how industry balances yield, speed and cost when choosing reaction conditions).
External Assessment Integration دمج التقييمات الخارجية	AP Chemistry Exam (College Board): Course and Exam Description Unit 7; AP-style lab and argumentation skills for the free-response section.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Self-management (preparing for the assessment and working safely in the lab), thinking (predicting and justifying shifts), communication (writing a clear reflection journal) and reflection (evaluating my own investigation).
Standards المعايير	• **7.7 Calculating Equilibrium Concentrations – TRA-7.E:** Identify the concentrations or partial pressures of chemical species at equilibrium based on the initial conditions and the equilibrium constant. • **7.9 Introduction to Le Châtelier's Principle – TRA-8.A:** Identify the response of a system at equilibrium to an external stress, using Le Châtelier's principle. • **7.10 Reaction Quotient and Le Châtelier's Principle – TRA-8.B:** Explain the relationships between Q, K, and the direction in which a reversible reaction will proceed to reach equilibrium.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Inquiry:** How do chemists control the conditions of a reaction to get more of the product they want? **TOK:** Le Châtelier's principle predicts what a system will do but does not explain why at the particle level. Is a rule that predicts well enough to count as scientific knowledge?
Success Criteria معايير النجاح **(Measurable)**	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Session 1 – 7.9 Le Châtelier's Principle** • **MUST (DOK 1):** I can state Le Châtelier's principle and identify the stress applied to a system at equilibrium. • **SHOULD (DOK 2):** I can predict the direction of the shift when concentration, volume or temperature changes, and explain why adding a catalyst causes no shift. • **COULD (DOK 3):** I can justify each shift by comparing Q with K, and explain why only a change in temperature changes the value of K. **Session 2 – Formative Assessment: ICE Tables and Le Châtelier** • **MUST (DOK 1):** I can set up an ICE table correctly and identify the direction of the shift for a given stress. • **SHOULD (DOK 2):** I can calculate equilibrium concentrations or K using an ICE table, and predict how the system responds to a change. • **COULD (DOK 3):** I can justify my answers using Q versus K and, after feedback, correct my errors and explain each misconception. **Session 3 – Lab: Equilibrium**

	• **MUST (DOK 1):** I can follow the lab procedure safely and record my observations (such as colour changes) in a clear table. • **SHOULD (DOK 2):** I can identify the stress in each test and the direction in which the equilibrium shifted. • **COULD (DOK 3):** I can justify each shift using Q versus K, and use my temperature results to decide whether the forward reaction is endothermic or exothermic. **Session 4 – Lab Reflection Journal (Summative Project)** • **MUST (DOK 1):** I can summarize the aim, method and main results of my lab. • **SHOULD (DOK 2):** I can explain my results using Le Châtelier's principle, with evidence from my observations and data. • **COULD (DOK 3):** I can evaluate the limitations and sources of error, propose specific improvements, and connect my findings to how industry controls reaction conditions.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Arabic Language اللغة العربية **Area:** Sustainable industry and energy in the UAE, linked to the UAE Net Zero by 2050 strategic initiative. **المجال** **Implementation:** **التطبيق** • Session 1: students apply Le Châtelier's principle to the reversible reaction that makes ammonia, as produced in Ruwais, and explain how pressure and temperature are chosen to increase the yield. • Session 4: in their reflection journal, students connect their lab findings to how UAE industry controls reaction conditions to save energy and reduce waste. • Arabic key terms: مبدأ لوشاتلييه (Le Châtelier's principle), إزاحة الاتزان (equilibrium shift), ثابت الاتزان (equilibrium constant), الضغط (pressure), درجة الحرارة (temperature).